6 Mechanistic Modelling of Recurrent Epidemics III; $\mathcal{R}_0$



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3
Mathematical Biology

Instructor: David Earn

Lecture 6
Mechanistic Modelling of Recurrent Epidemics III; $\mathcal{R}_0$
Tuesday 20 October 2026

Announcements

- Project Description Document is posted.
  - The project is to be submitted as a paper in the style of a research article for publication. It is *not* just a big assignment.
  - The project descriptions ask you questions and suggest directions for investigation, but you must construct the final document as a coherent single manuscript, not answers to a bunch of questions.
- **Midterm test:**
  - *Date:* Tuesday 24 November 2026 (tentative)
  - *Time available:* 2:30–5:20 p.m. (test duration: 90 minutes)
  - *Location:* in class, ETB 227
- **Assignment 3** is due Thursday 5 November 2026 at 11:59 p.m.
- **Assignment 4** is due Thursday 19 November 2026 at 11:59 p.m.
  - Make sure *you personally* can do the question on calculating $\mathcal{R}_0$ on this assignment *before* the midterm test.

$\mathcal{R}_0$: biological definition

The ***basic reproduction number*** $\mathcal{R}_0$ is:

> *the expected number of secondary cases produced, in a completely susceptible population, by a typical infective individual*

e.g., Anderson and May (1991) "Infectious Diseases of Humans"

$\mathcal{R}_0$: more mathematical definition

The ***basic reproduction number*** $\mathcal{R}_0$ is:

the number of new infections produced by a typical infective individual in a population at a disease free equilibrium (DFE)

van den Driessche and Watmough (2002) *Mathematical Biosciences* **180**, 29–48

$\mathcal{R}_0$: most mathematical definition

The ***basic reproduction number*** $\mathcal{R}_0$ is:

the spectral radius of the next generation operator at a disease free equilibrium (DFE)

Diekmann, Heesterbeek & Metz (1990) *J. Math. Biol.* **28**, 365–382

Definitions from matrix analysis

Definition (Spectrum of a matrix)

Let M be an $n \times n$ real (or complex) matrix. The ***spectrum of*** M is

$$\sigma(M) = \{\lambda : Mv = \lambda v \text{ for some non-zero } v \in \mathbb{C}^n\},$$

i.e., $\sigma(M)$ is the set of eigenvalues of M.

Definition (Spectral radius of a matrix)

Let M be an $n \times n$ real (or complex) matrix. The ***spectral radius of*** M is

$$\rho(M) = \max\{|\lambda| : \lambda \in \sigma(M)\},$$

i.e., $\rho(M)$ is the maximum modulus of the eigenvalues of M.

Computing $\mathcal{R}_0$

- In very simple models, $\mathcal{R}_0$ is the product of the transmission rate and the mean time in the infectious class. *e.g.*, In the SIR model with vital dynamics,

$$\mathcal{R}_0 = \beta \cdot \frac{1}{\gamma + \mu} .$$

- When there are multiple infected classes, it is more complicated to compute $\mathcal{R}_0$.
- In the SEIR model, we found (based on a biological argument) that

$$\mathcal{R}_0 = \beta \cdot \frac{\sigma}{\sigma + \mu} \cdot \frac{1}{\gamma + \mu} .$$

- Mathematically, the spectral radius of the next generation operator at the DFE is exactly this quantity. With this definition, it is also true that the disease persists if $\mathcal{R}_0 > 1$ and goes extinct if $\mathcal{R}_0 < 1$.

SEIR model (with vital dynamics)

$$\frac{dS}{dt} = \mu N - \frac{\beta SI}{N} - \mu S$$

$$\frac{dE}{dt} = \frac{\beta SI}{N} - \sigma E - \mu E$$

$$\frac{dI}{dt} = \sigma E - \gamma I - \mu I$$

$$\frac{dR}{dt} = \gamma I - \mu R$$

- Birth and death rate (μ)
- Transmission rate (β)
- Mean latent period ($1/\sigma$)
- Mean infectious period ($1/\gamma$)

Next generation matrix for the SEIR model

From now on, S, E, I, R denote population proportions.

- Consider flows in and out of the infected compartments, and highlight flows that correspond to new infections:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} E \\ I \end{pmatrix} = \begin{pmatrix} \beta SI - \sigma E - \mu E \\ \sigma E - \gamma I - \mu I \end{pmatrix}$$

- $\mathcal{F} =$ inflow of new infecteds to infected compartments $= \begin{pmatrix} \beta SI \\ 0 \end{pmatrix}$
- $\mathcal{V} =$ outflow from infected compartments minus inflow of *non-new* infecteds $= \begin{pmatrix} \sigma E + \mu E \\ -\sigma E + \gamma I + \mu I \end{pmatrix}$
- Let $F =$ linearization of $\mathcal{F}$ at DFE
- Let $V =$ linearization of $\mathcal{V}$ at DFE
- Then the ***next generation matrix*** is FV^{-1}
- Analogous to $\beta\gamma^{-1}$ in simple case.

Interpretation of FV^{-1} as next generation matrix

Almost verbatim from p. 33 of van den Driessche and Watmough (2002) *Mathematical Biosciences* **180**, 29–48

- To interpret the entries of FV^{-1} and develop a meaningful definition of $\mathcal{R}_0$, consider the fate of an infected individual introduced into compartment k of a disease free population.
- The (i, j) entry of F is the rate at which infected individuals in compartment j produce new infections in compartment i.
- The (j, k) entry of V^{-1} is the average length of time this individual spends in compartment j during its lifetime, assuming that the population remains near the DFE and barring reinfection.
- Hence, the (i, k) entry of the product FV^{-1} is the expected number of new infections in compartment i produced by the infected individual originally introduced into compartment k.
- Following Diekmann et al. (1990), we call FV^{-1} the next generation matrix for the model and set

$$\mathcal{R}_0 = \rho(FV^{-1}),$$

where $\rho(A)$ denotes the spectral radius of a matrix A.

$\mathcal{R}_0$ via FV^{-1} for the SEIR model

$$\mathcal{F} = \begin{pmatrix} \beta SI \\ 0 \end{pmatrix} \qquad \mathcal{V} = \begin{pmatrix} \sigma E + \mu E \\ -\sigma E + \gamma I + \mu I \end{pmatrix}$$

$$F = \begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix} \qquad V = \begin{pmatrix} (\sigma+\mu) & 0 \\ -\sigma & (\gamma+\mu) \end{pmatrix}$$

$$V^{-1} = \begin{pmatrix} \frac{1}{\sigma+\mu} & 0 \\ \frac{\sigma}{(\sigma+\mu)(\gamma+\mu)} & \frac{1}{\gamma+\mu} \end{pmatrix} \implies FV^{-1} = \begin{pmatrix} \frac{\beta\sigma}{(\sigma+\mu)(\gamma+\mu)} & \frac{\beta}{\gamma+\mu} \\ 0 & 0 \end{pmatrix}$$

$$\mathcal{R}_0 = \rho(FV^{-1}) = \frac{\beta\sigma}{(\sigma+\mu)(\gamma+\mu)}$$

- Note wrt previous slide that the $(2,1)$ entry of V^{-1} is the average time an individual who enters the E compartment spends in the I compartment: only a proportion $\sigma/(\sigma+\mu)$ of such individuals make it to the I compartment, where the average time spent—by individuals who get there—is $1/(\gamma+\mu)$.

Computing $\mathcal{R}_0$ for other compartmental ODE models

- The method applied in the previous slides to obtain $\mathcal{R}_0$ for the SEIR model works more generally for a very large class of "reasonable" infectious disease ODE models. For twice continuously differentiable vector fields, "reasonable" means:
  1. The vector field can be written $\mathcal{F} - \mathcal{V}$, where $\mathcal{F} \geq 0$ corresponds to *new infections* and $\mathcal{V}$ can be written $\mathcal{V} = \mathcal{V}^+ - \mathcal{V}^-$, where $\mathcal{V}^+ \geq 0$ corresponds to *outflow* and $\mathcal{V}^- \geq 0$ corresponds to *inflow of infectives that are not new*.
  2. The biologically relevant part of the state space is forward-invariant. In particular, if a compartment is empty, then there can be no transfer of individuals out of the compartment by death, infection, nor any other means.
  3. With new infection suppressed, all eigenvalues of the Jacobian at the DFE have negative real parts (if there is more than one DFE, $\mathcal{R}_0$ may depend on which one we focus on).
  4. Disease-free populations remain disease-free. New-infection terms enter only infected compartments.

Population size and the threshold theorem

- The theorem does not require a constant population size.
- It does require a disease-free equilibrium about which we can linearize.
- For a constant population, changing from counts to proportions preserves $\mathcal{R}_0$.
- For a changing population, a threshold for growth of the *infected proportion* need not be the threshold for growth of the *number infected*.

van den Driessche and Watmough (2002), Section 6, pp. 46–47

Computing $\mathcal{R}_0$ for other compartmental ODE models

Theorem (van den Driessche and Watmough (2002))

If the vector field associated with an ODE infectious disease model satisfies the conditions specified earlier, then

1. *$\mathcal{R}_0$ can be computed as $\rho(FV^{-1})$;*
2. *if $\mathcal{R}_0 < 1$ then the disease-free equilibrium (DFE) is locally asymptotically stable (LAS), whereas if $\mathcal{R}_0 > 1$ then the DFE is unstable.*

If $\mathcal{R}_0 > 1$ is sufficiently close to one, then under technical conditions that are often satisfied but awkward to state, there is also a LAS endemic equilibrium.

One susceptible class: model structure

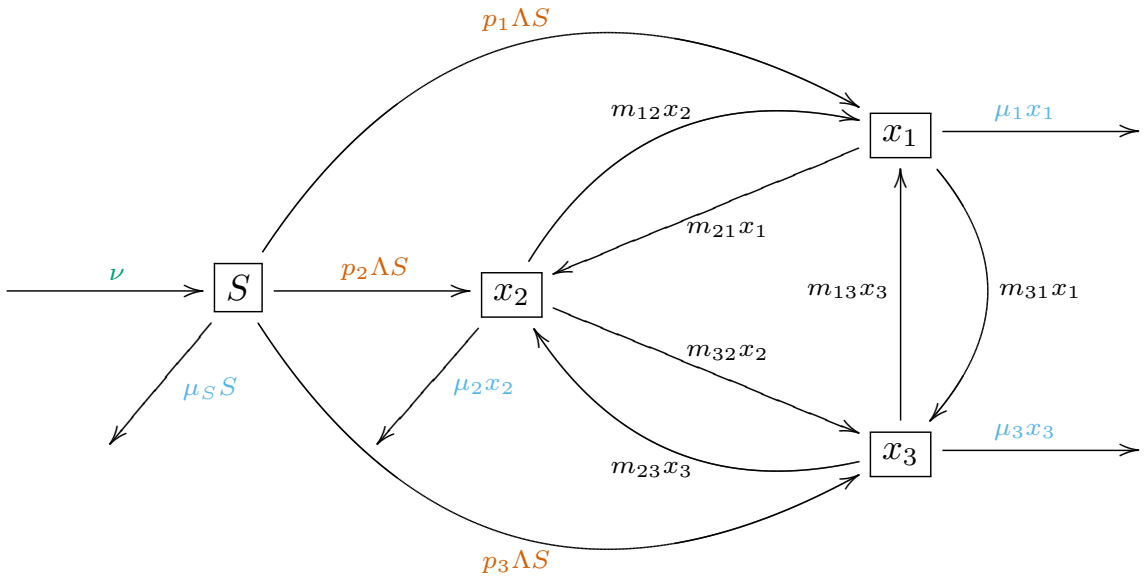


Fig. 1. Transfer diagram for the most general $Sx_1 \cdots x_n$ model with $n = 3$ nonsusceptible compartments (x_i). The subgraph excluding S is a complete directed graph on n nodes. There is no flow from any x_i back to S. The general class of $Sx_1 \cdots x_n$ models is defined formally by Eqs. 2–6.

S, x_i: numbers of susceptible and nonsusceptible individuals.
ν: recruitment into S; μ_S, μ_i: per capita mortality rates.
$\Lambda = \sum_j \beta_j x_j$: force of infection; β_j: transmission coefficient for x_j.
p_i: probability that a new infection enters x_i ($p_i \geq 0$, $\sum_i p_i = 1$).
m_{ij}: per capita transfer rate from x_j to x_i ($i \neq j$).

Earn DJD, McCluskey CC (2025), *PNAS* **122**(49):e2510156122. doi:10.1073/pnas.2510156122.

One susceptible class: global stability

In the paper's single-S model class: no reinfection or return to S, mass-action transmission and linear post-infection transfers.

Theorem (Earn and McCluskey (2025))

- *If $\mathcal{R}_0 \leq 1$, the DFE is GAS.*
- *If $\mathcal{R}_0 > 1$, the EE is GAS.*

GAS for all $\mathcal{R}_0 > 1$, not just LAS near one.

Eq. [7] gives, in the paper's count notation,

$$\boxed{\mathcal{R}_0 = \frac{\nu}{\mu_S}\,\vec{\beta}^{\mathsf{T}} M^{-1}\vec{P}}.$$

ν: susceptible recruitment; μ_S: susceptible per capita death rate;
$\vec{\beta}$: transmission coefficients; $\vec{P}$: new-infection entry probabilities;
M: post-infection transfer matrix (outflow minus inflow, including deaths).

Earn DJD, McCluskey CC (2025), *Global stability of epidemic models with uniform susceptibility*. *PNAS* **122**(49):e2510156122. doi:10.1073/pnas.2510156122.

$\mathcal{R}_0$ calculation: summary

- The biological method of deriving $\mathcal{R}_0$ is generally more informative in terms of what is going on. But it can be challenging to apply to complex models.
- The formal approach, *i.e.*, $\mathcal{R}_0 = \rho(FV^{-1})$, works in almost any situation you will encounter, even very complicated models with many compartments.
- If possible, it is best to use both methods to find an expression for $\mathcal{R}_0$, and make sure they agree.
- A completely different challenge is to estimate $\mathcal{R}_0$ for a real epidemic from data...

Estimating $\mathcal{R}_0$ based on the SEIR model

- If the SEIR model captures the natural history of some disease well, how can you *estimate* $\mathcal{R}_0 = \dfrac{\beta\sigma}{(\sigma+\mu)(\gamma+\mu)}$?
  - Mean latent period $1/\sigma$
  - Mean infectious period $1/\gamma$
  - Birth rate μ
  - Estimate β via initial growth rate r:
    - For the simplest SIR model, $r = \beta - \gamma$ so $\beta = r + \gamma$.
    - More generally, r is the largest positive (or least negative) real part of the eigenvalues of $F - V$.
    - For SEIR model we find:

$$r = \frac{1}{2}\left(\sqrt{4\beta\sigma + (\gamma-\sigma)^2} - (\gamma+\sigma+2\mu)\right)$$

    - Solving this for β we obtain: $\beta = \dfrac{(r+\sigma+\mu)(r+\gamma+\mu)}{\sigma}$

Estimating $\mathcal{R}_0$ via the renewal equation

Many epidemic models can be expressed with an integro-differential equation known as the ***renewal equation***, which relates the ***susceptible proportion*** $S(t)$ to the ***incidence*** $\iota(t)$,

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \mu - \iota(t) - \mu S(t)$$

$$\iota(t) = \mathcal{R}_0 S(t) \int_0^\infty \iota(t-s) g(s)\,\mathrm{d}s$$

where $g(s)$ is the ***intrinsic generation interval distribution***.

- ***generation interval*** = time difference between the moment when a focal individual was infected and the earlier time when the infector was infected

How can we use the renewal equation to obtain a formula that allows us to estimate $\mathcal{R}_0$?

- Set $S = 1$ and assume incidence is exponentially growing.

Estimating $\mathcal{R}_0$ from initial growth

- Near the fully susceptible DFE, setting $S = 1$ yields the ***linearized renewal equation***. Write $\iota(t) = \iota_0 e^{rt}$ for exponentially growing incidence.
- Substituting gives

$$\iota_0 e^{rt} = \mathcal{R}_0 \int_0^\infty \iota_0 e^{r(t-s)} g(s)\,\mathrm{d}s.$$

- Cancel $\iota_0 e^{rt}$ to obtain the ***Euler–Lotka equation***:

$$1 = \mathcal{R}_0 \int_0^\infty e^{-rs} g(s)\,\mathrm{d}s, \qquad \boxed{\mathcal{R}_0 = \frac{1}{\displaystyle\int_0^\infty e^{-rs} g(s)\,\mathrm{d}s}}$$

- Thus an estimate of the ***initial growth rate*** r, together with the ***generation-interval distribution*** g, gives an estimate of the ***basic reproduction number*** $\mathcal{R}_0$.

Wallinga J, Lipsitch M (2007), *How generation intervals shape the relationship between growth rates and reproductive numbers*. *Proc. R. Soc. B* **274**:599–604. doi:10.1098/rspb.2006.3754.

How can we derive the standard SIR model from the renewal equation?

SIR: infectious survival and generation intervals

- First consider the basic SIR model, without births or deaths ($\mu = 0$). The recovery rate is γ.
- If T is the infectious period, then

$$\Pr(T > s) = e^{-\gamma s}.$$

  This is the probability that an individual remains infectious s time units after infection.

- With constant infectivity while infectious, the generation-interval density is proportional to this survival probability:

$$g(s) = \frac{e^{-\gamma s}}{\int_0^\infty e^{-\gamma u}\,\mathrm{d}u} = \gamma e^{-\gamma s}.$$

- For this model, g is also the infectious-period density.

SIR prevalence from past incidence

- The population fraction infected between $t-s$ and $t-s+\mathrm{d}s$ is $\iota(t-s)\,\mathrm{d}s$.
- The proportion of that cohort still infectious at t is $e^{-\gamma s}$. Adding the surviving contributions from all past infections gives

$$I(t) = \int_0^\infty \iota(t-s)e^{-\gamma s}\,\mathrm{d}s.$$

- Since $g(s) = \gamma e^{-\gamma s}$, the renewal equation becomes

$$\begin{aligned}\iota(t) &= \mathcal{R}_0 S(t) \int_0^\infty \iota(t-s)\gamma e^{-\gamma s}\,\mathrm{d}s \\ &= \mathcal{R}_0 \gamma S(t) I(t) = \beta S(t) I(t),\end{aligned}$$

because $\mathcal{R}_0 = \beta/\gamma$.

Differentiating the prevalence integral

Differentiating with respect to t gives

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \int_0^\infty \iota'(t-s)e^{-\gamma s}\,\mathrm{d}s.$$

Since $\iota'(t-s) = -\frac{\mathrm{d}}{\mathrm{d}s}\iota(t-s)$, integration by parts gives

$$\begin{aligned}\frac{\mathrm{d}I}{\mathrm{d}t} &= \left[-\iota(t-s)e^{-\gamma s}\right]_{s=0}^{s=\infty} - \gamma\int_0^\infty \iota(t-s)e^{-\gamma s}\,\mathrm{d}s \\ &= \iota(t) - \gamma I(t) \\ &= \beta S(t)I(t) - \gamma I(t).\end{aligned}$$

The endpoint at $s=\infty$ is zero for bounded past incidence. The endpoint at $s=0$ supplies the new infections $\iota(t)$.

The basic SIR model recovered

Using $\iota = \beta SI$ in $\frac{\mathrm{d}S}{\mathrm{d}t} = -\iota$, and setting $R = 1 - S - I$, gives

$$\frac{\mathrm{d}S}{\mathrm{d}t} = -\beta SI, \qquad \frac{\mathrm{d}I}{\mathrm{d}t} = \beta SI - \gamma I, \qquad \frac{\mathrm{d}R}{\mathrm{d}t} = \gamma I.$$

In infectious-period time units, $\tau = \gamma t$ and $\alpha = \gamma s$,

$$\frac{\mathrm{d}S}{\mathrm{d}\tau} = -\mathcal{R}_0 SI, \qquad \frac{\mathrm{d}I}{\mathrm{d}\tau} = \mathcal{R}_0 SI - I,$$

and the generation-interval density becomes

$$g_\tau(\alpha) = \frac{1}{\gamma} g(\alpha/\gamma) = e^{-\alpha}.$$

This dimensionless exponential kernel is used in:

Earn DJD, Parsons TL, *Epidemic "momentum" and a conservation law for infectious disease dynamics*, arXiv, 7 September 2026. doi:10.48550/arXiv.2511.01939. Appendix A, pp. 19–20, Eqs. (A1)–(A3).

SIR with vital dynamics

- With recovery at rate γ and death at rate μ, the probability of remaining infectious for s time units is $e^{-(\gamma+\mu)s}$.
- Hence

$$g(s) = (\gamma+\mu)e^{-(\gamma+\mu)s}, \qquad \mathcal{R}_0 = \frac{\beta}{\gamma+\mu}.$$

- The same calculation gives $\iota = \beta SI$ and $\frac{\mathrm{d}I}{\mathrm{d}t} = \iota - (\gamma+\mu)I$.
- Combining this with the susceptible equation on the renewal slide recovers

$$\begin{aligned}
\frac{\mathrm{d}S}{\mathrm{d}t} &= \mu - \beta SI - \mu S,\\
\frac{\mathrm{d}I}{\mathrm{d}t} &= \beta SI - (\gamma+\mu)I,\\
\frac{\mathrm{d}R}{\mathrm{d}t} &= \gamma I - \mu R.
\end{aligned}$$

Technical statement of the $\mathcal{R}_0 > 1$ theorem of van den Driessche and Watmough

Endemic equilibria near $\mathcal{R}_0 = 1$

- Let $x' = f(x, \eta)$, where $x \in \mathbb{R}^n$ and the DFE x_* satisfies $f(x_*, \eta) = 0$ for all nearby η. Choose η so that $\mathcal{R}_0 < 1$ for $\eta < 0$ and $\mathcal{R}_0 > 1$ for $\eta > 0$.
- At $\eta = 0$, suppose $\mathcal{R}_0 = 1$ and $A = D_x f(x_*, 0)$ has a simple zero eigenvalue, with all other eigenvalues having negative real parts.
- Choose right and left nullvectors w, v with $Aw = 0$, $v^T A = 0$, $v^T w = 1$, and non-negative infected components.

Evaluating all derivatives at $(x_*, 0)$, define

$$a = \frac{1}{2} \sum_{i,j,k=1}^{n} v_i w_j w_k \frac{\partial^2 f_i}{\partial x_j \partial x_k},$$

$$b = \sum_{i,j=1}^{n} v_i w_j \frac{\partial^2 f_i}{\partial x_j \partial \eta}.$$

Endemic equilibria near $\mathcal{R}_0 = 1$

Theorem (van den Driessche and Watmough (2002), Theorem 4)

Assume the epidemic-model conditions and the setup on the previous slide, with f twice continuously differentiable in x and η, and $b \neq 0$. Then there is a $\delta > 0$ such that

1. *if $a < 0$, there are LAS endemic equilibria near x_* for $0 < \eta < \delta$, where $\mathcal{R}_0 > 1$;*
2. *if $a > 0$, there are unstable endemic equilibria near x_* for $-\delta < \eta < 0$, where $\mathcal{R}_0 < 1$.*

These are, respectively, ***forward*** and ***backward bifurcations***.

The stable-endemic-equilibrium conclusion is local: $\mathcal{R}_0$ must be sufficiently close to one, and $a < 0$.

van den Driessche and Watmough (2002), pp. 42–44